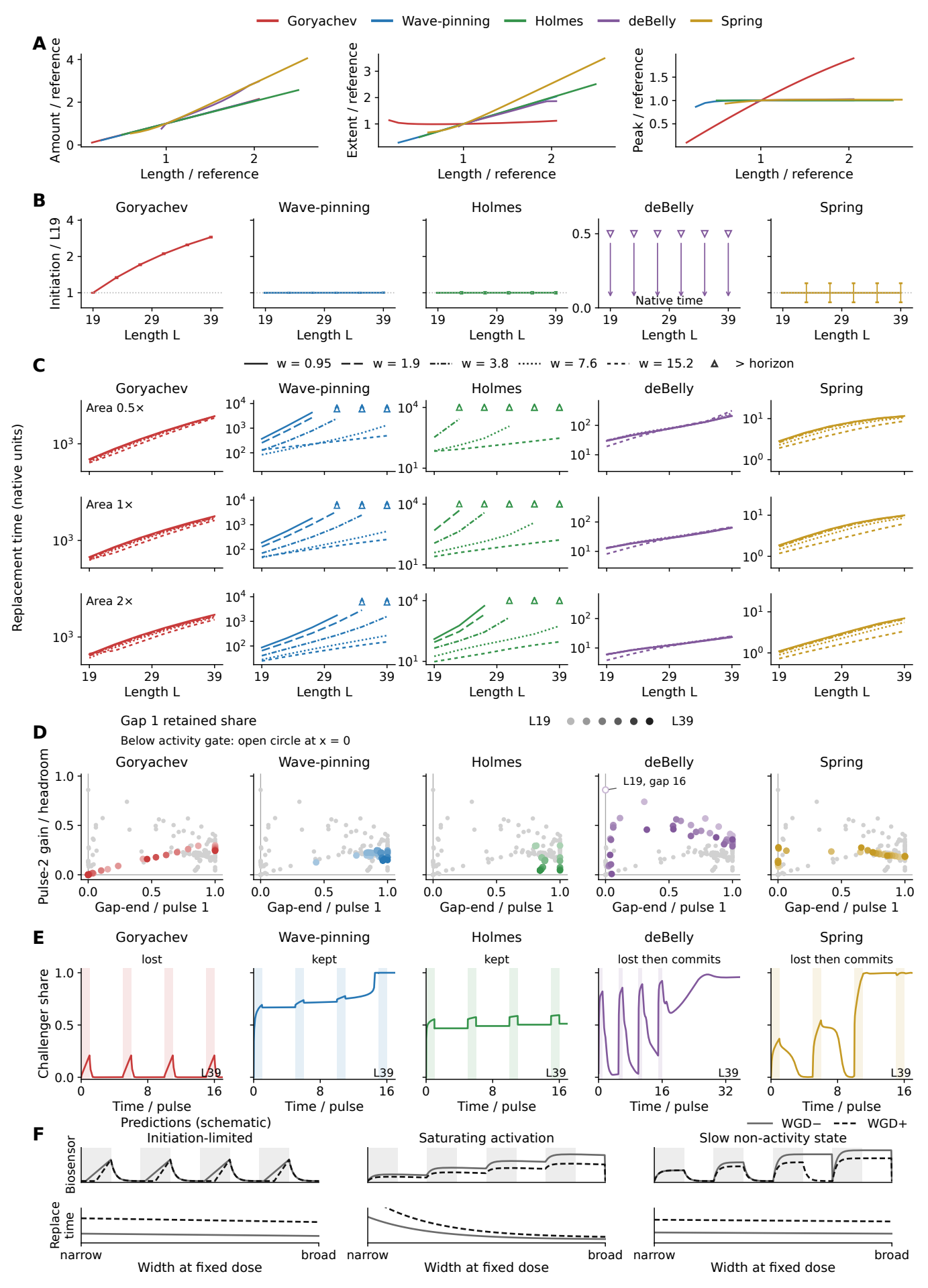

Goryachev
Wave-pinning
Holmes
deBelly
Spring
A
Amount / reference
Length / reference
Extent / reference
Peak / reference
B
Initiation / L19
Native time
Length L
C
w = 0.95
w = 1.9
w = 3.8
w = 7.6
w = 15.2
> horizon
Area 0.5×
Area 1×
Area 2×
Replacement time (native units)
D
Gap 1 retained share
Below activity gate: open circle at x = 0
L19
L39
L19, gap 16
Pulse-2 gain / headroom
Gap-end / pulse 1
E
lost
kept
lost then commits
Challenger share
Time / pulse
F
Predictions (schematic)
WGD−
WGD+
Initiation-limited
Saturating activation
Slow non-activity state
Biosensor
Replace time
narrow
broad
Width at fixed dose

Separation slows the resolution of coexisting regions at short range; at matched separation, cell size dominates both driven takeover and resolution.

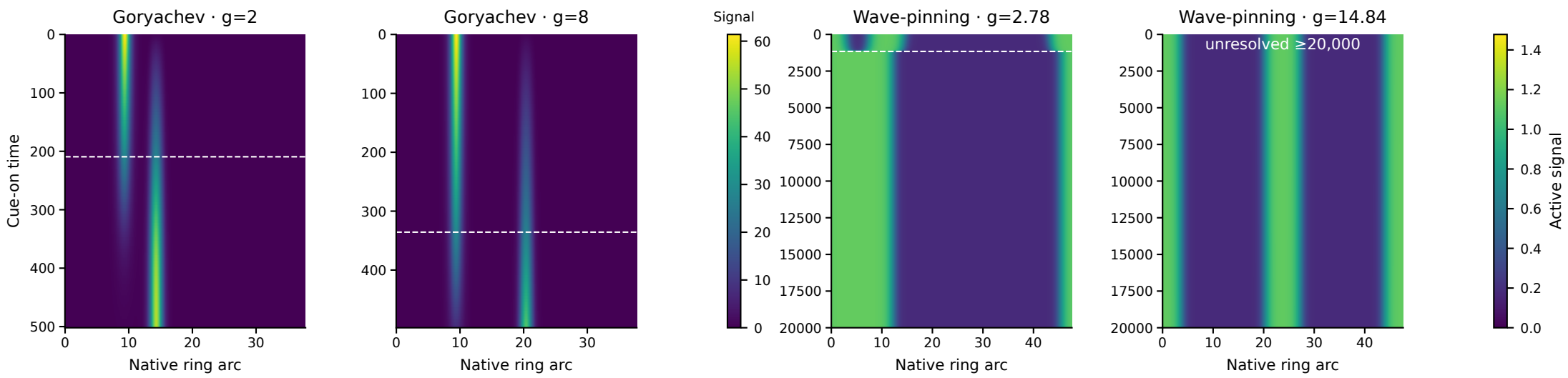
A Rotating a cue separates distance from native cell size



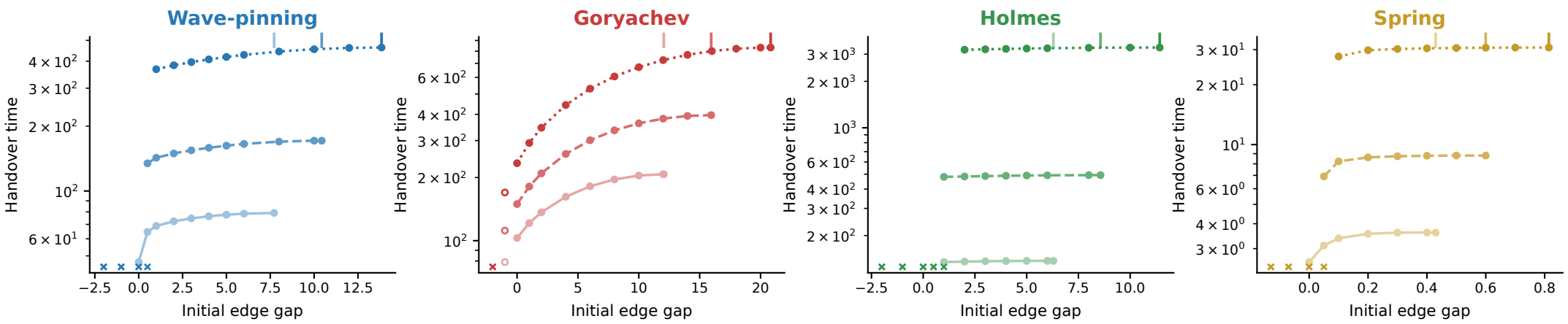
Same physical gap across sizes

A1: continuous cue → handover
B: equal complete-region amounts,
cue withdrawal → single-peak resolution
Blue incumbent; orange cue; g is edge gap.

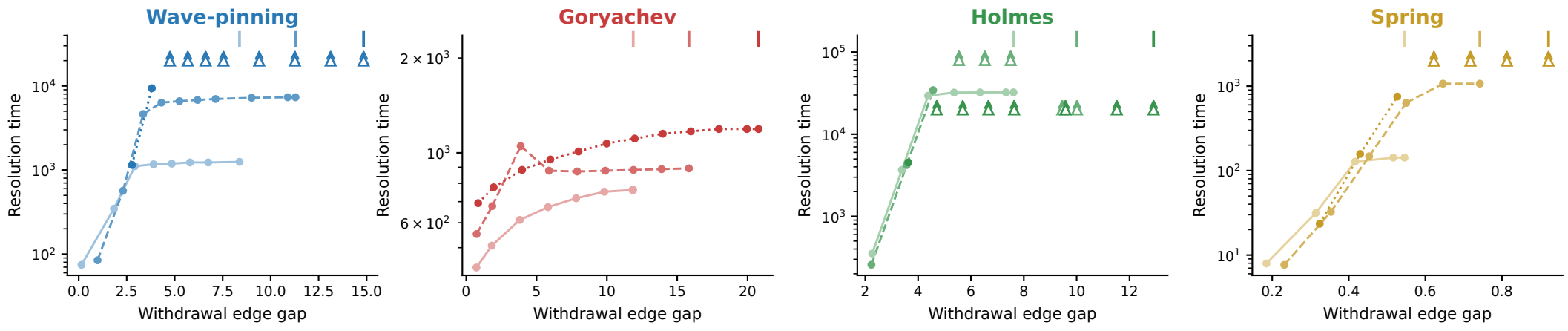
B Native histories: driven handover (left); time after withdrawal (right)



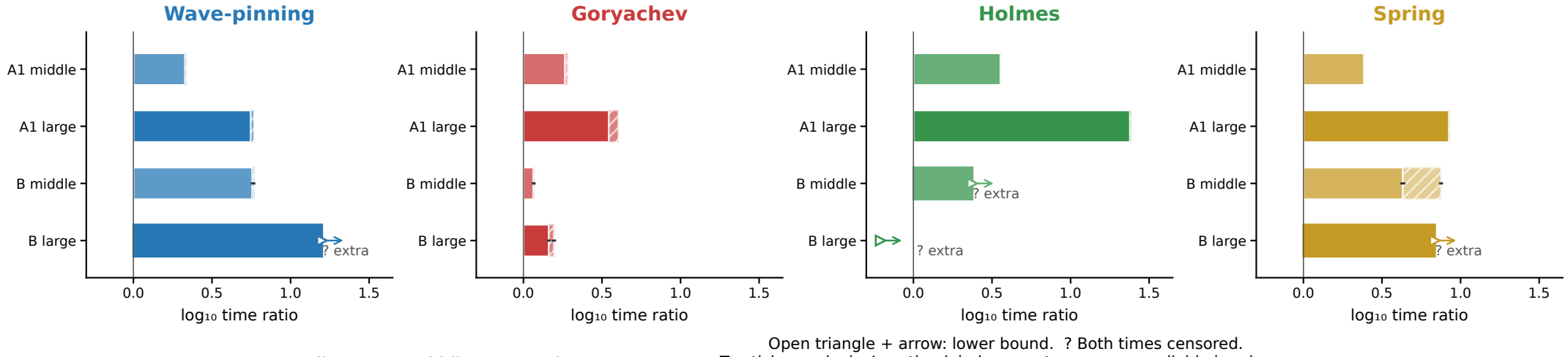
C Driven handover against initial edge gap



D Cue-free resolution against measured withdrawal gap · equal amounts



E Size at matched gap (solid) + extra separation (hatched)

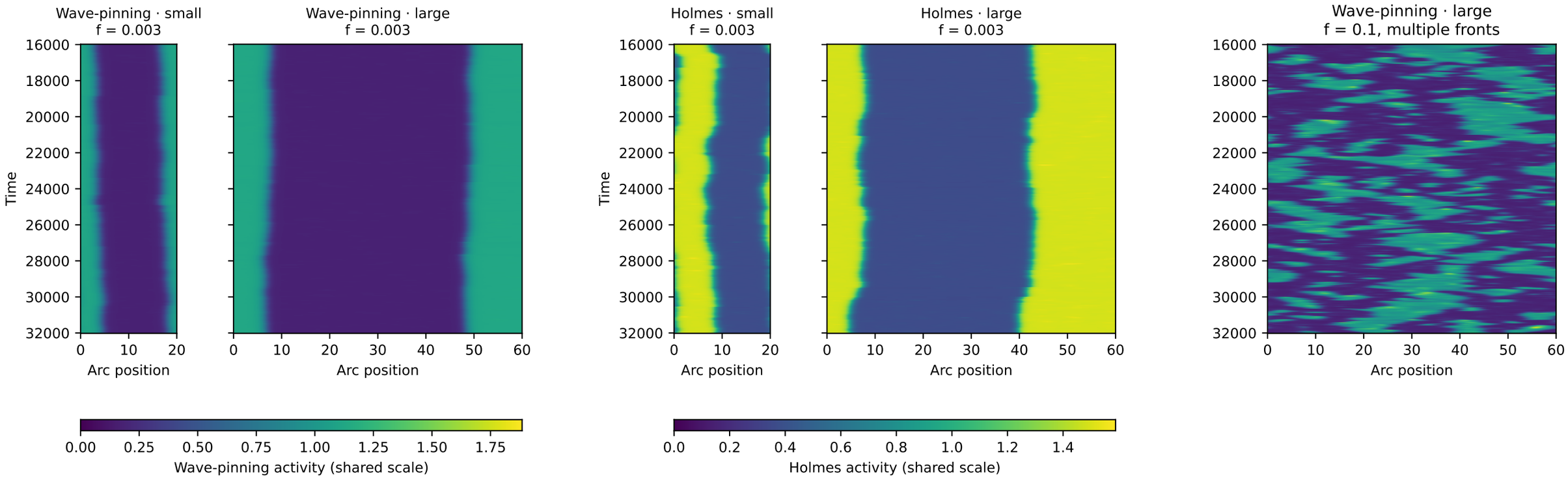


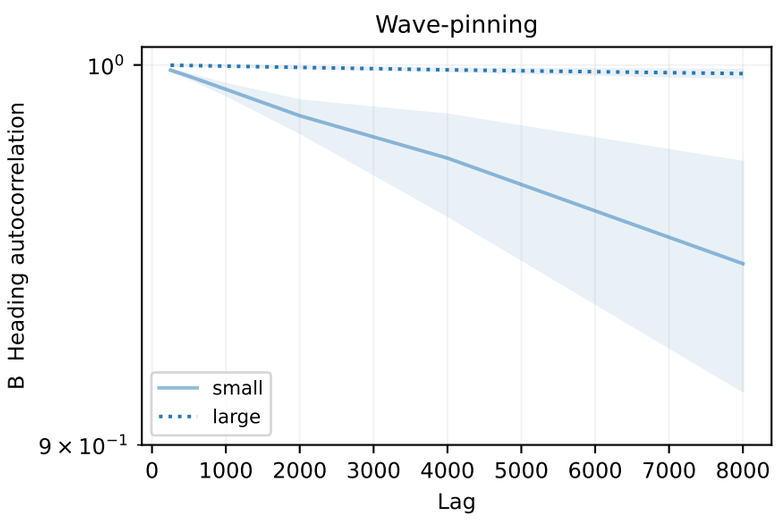
Open triangle + arrow: lower bound. ? Both times censored.
Top ticks: each size's antipodal-placement gap. ×: unavailable handover.

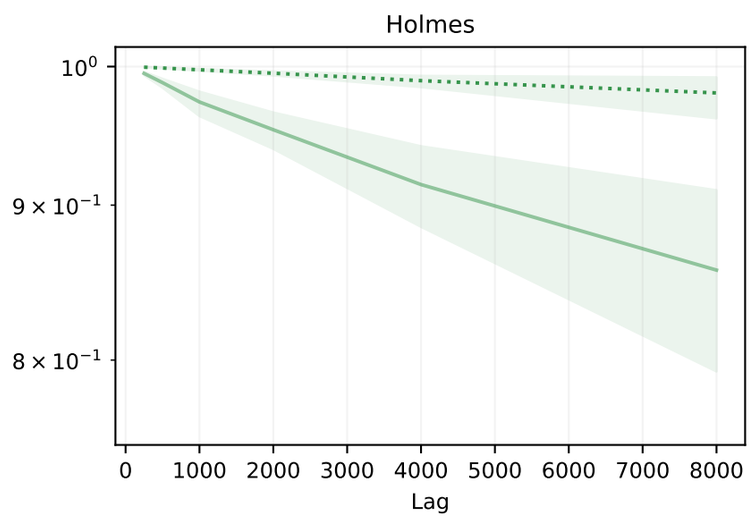
Enlarged cells hold heading longer under fluctuating input

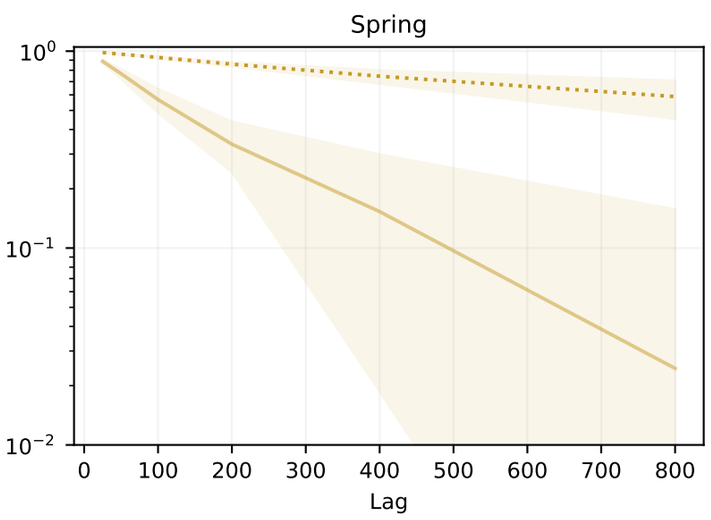
Gains reach or exceed geometry at each model's weakest measurable input; the advantage declines as input strengthens

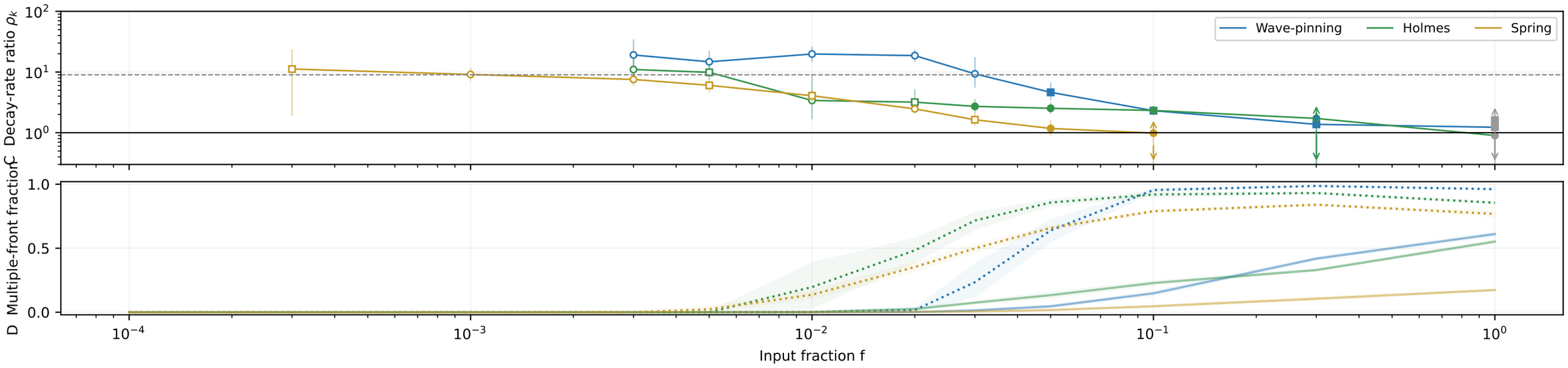
A Native activity histories

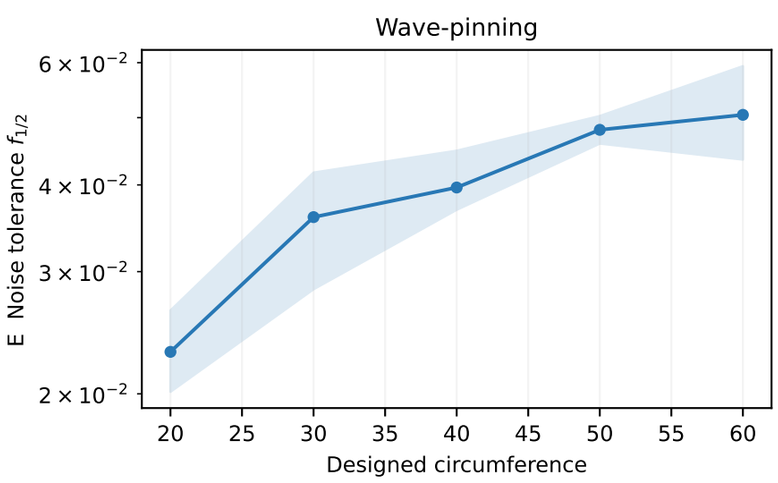


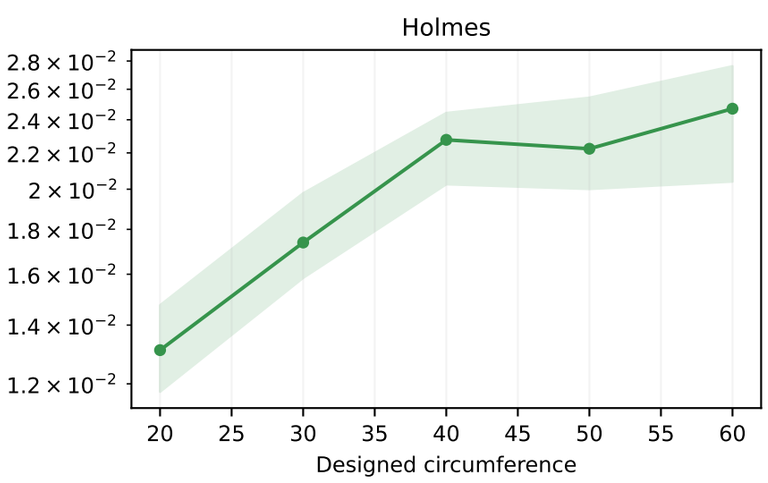


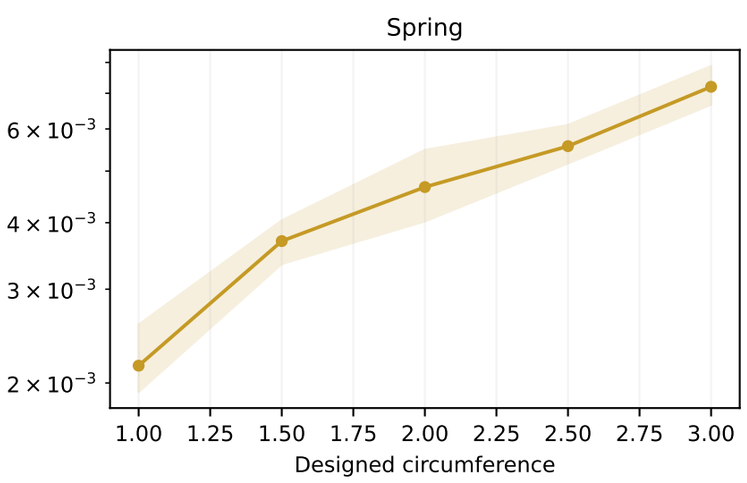












Small: lighter solid; large: darker dotted. C: open fill single front; filled multiple fronts; grey fill low contrast; square >1.5× lag drift.
Pointwise whole-seed 95% CIs. Horizontal references: ratio 1 and nominal geometry 9. D is descriptive; it does not establish a cause of erosion.
H1 uses a post hoc f < large-cell f½ domain; frozen H1 remains inconclusive. Goryachev retained in supplement only.